Before: [1, 1, 2, 3, 3, 8, 9, 9, 10]

After: [1 2 3 8 9 10 — — —]

return = 6

[1, 2, 3, 8, 8, 9, 9, 10, 9, 9, 10] T

responsible for finding the next pos.

[1, 2, 3, 8, 8, 9, 9, 10, 9, 9, 10] T

responsible for finding the next uniq element.

[1, 1, 1, 2, 3, 3, 3, 8, 9, 9, 9, 10].

Hashing.

0 → hello.

15 → gfg.

12 → apple.

13 → car.

19 → bike.

21 → cycle

0	12	13	19	15							21
hello	—	—	car	bike	gfg	—	—	—	—	—	cycle

Dict.

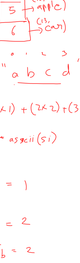
Sparse nature of the array.

TC.

Insert : O(1)

Delete : O(1)

Get : O(1)



(key mod 7) HF.

Key → Hashing function → Gives a valid index in the Hash Table.

Collision: 2 or more keys map to the same index in the Hash Table.

Key = 19.

19 → $\boxed{\text{int}}$ → 0

" a b c d "

" a a f "

" f o "

(0×0) + (1×1) + (2×2) + (3×3)

$\sum i^{\text{th}}$ ascii(s[i])

a e

a b

a c

d b

d c

⋮

a b = 1

a c = 2

a a b = 2

$$\boxed{\sum \text{ascii}(s[i]) \times 37^{i}}$$

$a_{\underline{c}}$ ^ $37^{\underline{i}}$

2 1

1 2